



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Preparing Yourself to Understand Data Migration



Learning Outcome:

Implement data migration processes and data backups within the IT solution.
Understand the different types of data movements, making an emphasis on data migration movements as well as data backups.

Ask your partner:

- What are the most significant advantages of data migration?
- What is the importance of data migration for regular people like students, business people or parents?

Can Do Statements:

- I can talk about common issues that can cause problems in the data migration process.
- I can describe array-to-array data migrations.
- I can explain file-to-file data migrations process.





Let's Explore!

Two students are checking information that will help them with their presentation about data migration.



Instructions: Read the text below.

Mike: Hello Dave. What are you reading?

Dave: I'm reading an interesting article that might help us with our presentation next week.

Mike: The one about data migration? What does it say?

Dave: Well. It says "If you're planning to run a data migration project, you will need to bear in mind some common issues regarding that topic. The most common problems are the complexity of those projects. Different types of storage and files, even different servers make of migration a task of high complexity. Operational dependencies represent a huge issue because IT departments work 24/7, which makes it difficult to meet the conditions to migrate data without outage. Technical dependencies make such projects run slow and increase the cost because some software and hardware need to be updated before the data migration. Scale is also important since large amounts of files make the process take longer than expected. All those factors increase the cost of the project as well. Migration demands a huge investment and sometimes such projects can make people feel tired. However, data migration is something necessary and once it's achieved, it's really satisfying."

Mike: Wow, data migration might be discouraging and daunting. Anyway, does it mention anything about array-to-array transfer?

Dave: Yes, it does. To sum up, there are host based, array based, hypervisor based, virtualization appliance based, and virtual array based. The last ones allow seamless data migration.

Mike: Even though the migration process makes you feel exhausted, it is also fulfilling to know everything works well after such effort. What about file-to-file migration?

Dave: It only says that if people want to perform file-to-file migration, they will need to understand that it's equally complex. Making data available during and after the migration process without affecting the location or address of the files is something discouraging.





Let's Practice!

Mike is getting ready for the presentation.

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: It will be / have problems / migrate data / If you want / A take longer to finish.

1. _____ to migrate data, you will be facing complexity problems.
2. If the amount of information is large, it will _____.
3. You will _____ when migrating information if the files format are different.
4. If you want to _____, it will be necessary to update hardware and software.
5. _____ more expensive to migrate data if you have to update your software and hardware.



Let's Practice!

Mike is reviewing file to file data migration.

Instructions: Read the text again. Select the correct answer.

1. Why is file to file data migration difficult?
 - a. Because it is boring.
 - b. Because the data needs to be available.
 - c. Because it takes a lot of time.
 - d. Because it is difficult to do.
2. Why does it take so much time to migrate data with the file-to-file method?
 - a. Because you need to create copies of all the data.
 - b. Because only two people do the job.
 - c. Because is very difficult to understand.
 - d. Because you need many people to do it.
3. Why do you need copies of all the information?
 - a. Because you need 3 copies of everything.
 - b. In case the person migrating data gets sick.
 - c. In case the data is misplaced or deleted.
 - d. Because the regulations say so.
4. What is a data migration plan?
 - a. It is a way to know who did what and pay accordingly.
 - b. Something you do to migrate data in an organized and efficient way.
 - c. A plan to be sure who is making mistakes and fired that person.
 - d. It is a plan for the future of data.
5. What is other problem of file to file data migration?
 - a. It has to be done by a computer.
 - b. It is impossible to do it.
 - c. It has to be done manually.
 - d. Many people are needed to do this.
6. What do people who the perform file-to-file method have to do?
 - a. They need to understand that it's a complex process.
 - b. They need to know that it's a very simple process.
 - c. They need to understand that it's simple.
 - d. They need to learn the information by heart.





Extension Activity

Plan a data migration process.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Imagine you work for a company that needs to make a data migration process.
- Create a plan to make the data migration. Be specific, tell which type of processes you would use and why.
- Make a document with your plan.
- Present the plan to a partner and ask his/her question about it.

Self-Assessment

Yes No

- I create a plan for a data migration process. ☐ ☐
- I create a plan in which I explain every step of the process I would follow. ☐ ☐
- I make a document explaining my plan. ☐ ☐
- I present my plan to a partner and answer questions about it. ☐ ☐



Target Words

- **outage** (noun): an intentional and temporary suspension of activities or operation.
e.g. Operational dependencies represent a huge issue because IT departments work 24/7, which makes it difficult to meet the conditions to migrate data without outage.
- **dependency** (noun): the feeling of reliance on something or somebody.
e.g. Technical dependencies make such projects run slow and increase the cost because some software and hardware need to be updated before the data migration.
- **daunting** (adjective): discouraging.
e.g. Wow, data migration might be discouraging and daunting.
- **sum up** (verb): to synthesize or condense the most relevant aspects of a topic.
e.g. To sum up, there are host based, array based, hypervisor based, virtualization appliance based, and virtual array based.
- **virtualization** (noun): a simulation done in a computer.
e.g. To sum up, there are host based, array based, hypervisor based, virtualization appliance based, and virtual array based.
- **seamless** (adjective): with no interruption.
e.g. The last ones allow seamless data migration.



Learning Points

Adjectives Ending in -Ed and -Ing

There are two different types of **adjectives**, the ones that end in **-ed** and the ones that end in **-ing**.

A suffix is a particle that we add at the end of a word. In this case, the suffix **-ed** and **-ing** help us create different adjectives with different meanings.

The adjectives that end in **-ed** help us describe emotions; in simple words, they describe the way someone feels about something.

For example:

*I was very **bored** in the technology class.*

*She was **surprised** by the way data migrates.*

*I feel **confused** about what migration technology I should use.*

On the other hand, the adjectives that end in **-ing** describe a situation.

For example:

*The technology class was very **boring**.*

*The way data migrates is **surprising**!*

*Data migration can seem **stressing** at first, but you just need to practice.*

Examples

- *I'm reading an interesting article that might help us with our presentation next week.*
- *However, data migration is something necessary and once it's achieved, it's really satisfying.*
- *Even though the migration process makes you feel exhausted, it is also fulfilling to know everything works well after such effort.*

Types of Data Migration

There are different types of **data migration** people can use to migrate files according to the different features a project has.

We can subdivide data migration in two categories: Array-to-array and file-to-file.

Additionally, the **array-to-array** can be divided into the following types:

Host based: Copying volumes at host level.

Array based: Moving data between arrays.

Hypervisor based: To move virtual machines between storage on varied arrays.

Virtualization appliance based: To abstract the storage from the host using an app. It allows seamless migration.

Virtual array based: Virtualizing external storage and moving data between arrays.

The other type is the **file-to-file** modality which is equally complex in order to make data available to users during and after the migration process, and not affecting the location or address of such files.

